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Term Project

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STORMWATER MANAGEMENT AT WILLOW ELEMENTARY SCHOOL, LAKEWOOD, CA

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Abstract

The stormwater drainage system at Willow Elementary School in Lakewood, California, experiences excessive surface runoff due to urbanization, posing risks of localized flooding and system overload. This study evaluates the current system and proposes a sustainable solution to reduce runoff volumes using porous pavements as a low-impact development (LID) strategy. The analysis was conducted using the Storm Water Management Model (SWMM) to simulate pre-development, post-development, and post-LID scenarios under various storm events, including 5-, 10-, 25-, and 50-year return periods.

Data inputs such as rainfall intensity, soil characteristics, land cover, and elevation were collected from reliable sources, including NOAA Atlas 14 and the Web Soil Survey. The SWMM simulations revealed significant increases in surface runoff in the post-development scenario compared to pre-development conditions. However, the implementation of porous pavements as an LID measure effectively decreased runoff volumes, mitigating flooding risks and improving system performance during moderate to severe storm events.

Results demonstrate a significant reduction in the runoff with the implementation of porous pavements. Surface runoff decreased by 43.6% for the 50-year event, 44.1% for the 25-year event, 44.7% for the 10-year event, and 45.4% for the 5-year event compared to post-development conditions. These findings highlight the efficacy of porous pavements in reducing urban runoff and enhancing stormwater management systems in highly developed areas. This study provides a practical framework for addressing similar challenges in Lakewood and contributes to advancing sustainable drainage solutions for urban environments.

Introduction

Willow Elementary School in Lakewood, California, has gone through significant development which has greatly disrupted the natural water cycle by replacing vegetated, permeable surfaces with impervious materials such as asphalt, concrete, and rooftops. These changes have resulted in excessive surface runoff during rainfall events, overwhelming the existing stormwater drainage system. The increased runoff contributes to localized flooding, soil erosion, and strain on infrastructure, posing safety risks and operational challenges. With the capacity of traditional stormwater systems unable to meet the growing demands of urban development, there is an urgent need for a sustainable and effective solution.

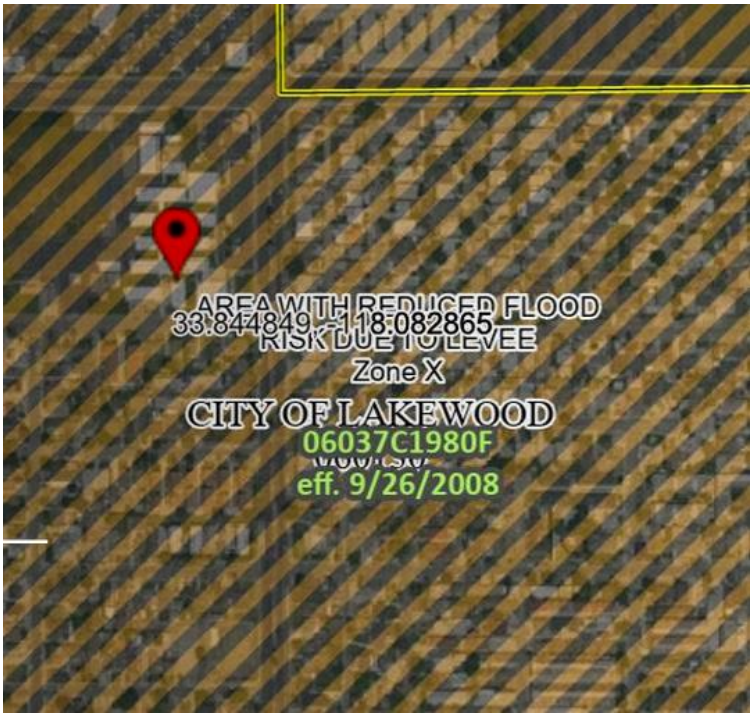

Figure 1: Map showing flood risk at Willow Elementary School (Source: FEMA)

This project investigates porous pavements as a low-impact development (LID) strategy to address excessive runoff in the Willow Elementary School neighborhood. Porous pavements enhance infiltration, reduce runoff, and alleviate pressure on drainage systems, offering an eco-friendly solution to urban stormwater challenges. Using the Storm Water Management Model (SWMM), developed by the EPA, this study evaluates the effects of porous pavements on runoff reduction and flood mitigation under various storm scenarios. The findings highlight their potential to improve drainage performance and resilience in urbanized areas like Lakewood, California.

Project Background

Location

Willow Elementary School, located at 11733 East 205th Street, Lakewood, California, is part of the ABC Unified School District. Serving approximately 388 students from kindergarten through sixth grade, the school is situated in a primarily residential neighborhood dominated by single-family homes and urban infrastructure. The area does feature adequate green spaces, but most of the landscape is covered by impervious surfaces, including roads, sidewalks, and parking lots. These surfaces significantly hinder natural infiltration, increasing surface runoff during rainfall events. The neighborhood's stormwater drainage system is connected to Lakewood's municipal stormwater management network which falls under the Los Angeles County Stormwater System, which is responsible for directing runoff to appropriate discharge points. However, like many urbanized areas, the existing infrastructure at and around Willow Elementary struggles to accommodate the increased runoff volumes generated by urban development.

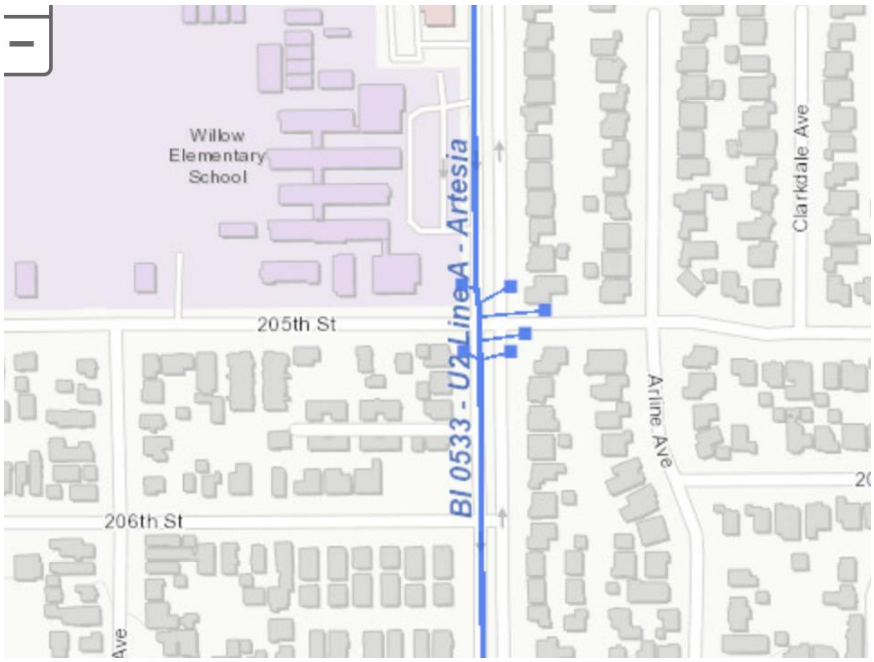

Figure 2: Stormwater drainage map of Willow Elementary School (Source: LA County Public Works)

History

Lakewood's stormwater management challenges date back to the rapid urbanization of the mid-20th century, which transformed the region from an agricultural hub into a thriving suburban community. A pivotal moment in the city's history occurred in 1952,

when a record-breaking flood overwhelmed the area with up to eight inches of rainfall in some parts of Los Angeles County. In response, the County of Los Angeles revamped its stormwater infrastructure, laying the groundwork for modern stormwater management systems. Despite these efforts, ongoing urbanization has led to increasing volumes of runoff, and overwhelming existing systems during peak rainfall events.

Figure 3: Flooding of Lakewood in 1952 (Source: USC Library)

In recent decades, Lakewood has adopted certain measures, including public education campaigns, infrastructure upgrades, and compliance with federal water quality regulations. The municipal separate storm sewer system (MS4) permit in Lakewood, regulates stormwater discharges under the Los Angeles Regional Water Quality Control Board, requiring watershed-based management, pollutant reduction programs, and public outreach to prevent stormwater pollution. However, localized flooding persists, particularly in neighborhoods surrounding Willow Elementary School. The school reflects the bigger challenges seen in cities, with its old drainage system struggling to manage the growing amount of water from urban areas. These conditions highlight the urgent need for sustainable solutions like LID to reduce runoff volumes and enhance stormwater management resilience.

Method of Analysis

Data Sources

To accurately simulate stormwater conditions at Willow Elementary School, reliable and authoritative data sources were utilized to define model parameters such as rainfall intensity, soil properties, elevation, land cover, and existing drainage infrastructure. These datasets were selected for their relevance to the study area and their ability to provide precise inputs for the Storm Water Management Model (SWMM). The table below lists the key data sources, along with the purpose of each dataset.

Data Type	Source	Purpose
Rainfall Data	NOAA Atlas 14	Provided historical and design storm rainfall values and analyzed in MS-Excel for equal time intervals.
Soil Properties	Web Soil Survey (USDA)	Supplied soil type and infiltration characteristics for the study area (Sandy loam soil).
Elevation Data	Google Earth	Determined elevation changes and slopes across the school property and surrounding areas.
Land Cover	Google Earth	Provided information on impervious and pervious surface percentages.
Existing Stormwater Infrastructure	Los Angeles County Public Works Storm Drain System	Mapped existing drainage system, conduit dimensions, and flow paths.
Porous Pavement Guidelines	U.S. Environmental Protection Agency (EPA)	Supplied design parameters and performance data for implementing porous pavements.

Data Analysis

To assess stormwater runoff conditions at Willow Elementary School, the Storm Water Management Model (SWMM) was used to simulate pre-development, post-development, and post-LID (porous pavement) conditions. The analysis has been done in metric units and involved processing key inputs, assumptions, and parameters, as outlined below:

1. Post-Development Scenario:
    - Modeled current urbanized conditions with increased impervious areas.
    - Impervious percentages ranged from 52.3% to 80%, with composite Curve Number (CN) values between 80.3 and 90.6.

Figure 4: SWMM (Post) model representation of all the Subcatchments

- Conduits were modeled as rectangular concrete boxes (1.8288 m depth, 3.048 m width) to simulate the existing drainage system. The junctions in the storm system (refer to Figure 4) are named sequentially from the lowest point in the system, starting with J1 at the bottom and progressing upward to J5 near Willow School, following the same pattern as the conduits.
- Time of Concentration (Tc) was calculated using the iterative equation shown in Figure 4. The values were iteratively adjusted until convergence was achieved, reflecting the subcatchment's specific flow conditions.

$$T_c = \frac{0.31 \times L^{0.483}}{(C_d \times I_t)^{0.519} \times S^{0.135}}$$

Figure 5: Equation used for the calculation of Time of Concentration

Subcatchment	Total Area(ha)	Area(m2)	L(m)	W(m)	Impervious Area(ha)	Percentage Impervious	Slope(%)	Composite CN	Drains to
S1	4.6	46000	309.94	148.416	2.405	52.289	0.10	80.34693	J5
S2	0.57	5700	177.5	32.113	0.456	80.000	0.17	90.6	J4
S3	0.69	6900	180	38.333	0.487	70.588	0.25	87.11756	J3
S4	0.73	7300	197.1	37.037	0.527	72.222	0.08	87.72214	J1
S5	0.77	7700	194	39.691	0.580	75.263	0.02	88.8467	J2
S6	1.7	17000	204.6	83.089	1.214	71.429	0.05	87.42812	J1

Junction	Invert(m)	Max Depth(m)
J1	7.57	1.8288
J2	7.56	1.8288
J3	7.46	1.8288
J4	7.45	1.8288
J5	7.44	1.8288

Conduit	Type	Max Depth(m)	Bottom Width(m)	Length(m)
C1	Rectangular Concrete Box	1.8288	3.048	5.43
C2	Rectangular Concrete Box	1.8288	3.048	27.43
C3	Rectangular Concrete Box	1.8288	3.048	5.76
C4	Rectangular Concrete Box	1.8288	3.048	10

Figure 6: SWMM (Post) Model Input

2. Pre-Development Scenario:
    - Represented natural conditions with minimal impervious areas.
    - Impervious areas were assumed to be 5% throughout the area, and Composite Curve Numbers (CN) were calculated as 73 for all subcatchments.
    - Slopes ranged from 0.02% to 0.25%, and drainage lengths and widths were calculated for flow paths which all led to one single outlet (J5).

Subcatchment	Total Area(ha)	Area(m2)	L(m)	W(m)	Impervious Area(ha)	Percentage Impervious	Slope(%)	Composite_CN	Drains to
1	4.6	46000	309.94	148.416	0.230	5.0	0.10	73	J5
2	0.57	5700	177.5	32.113	0.029	5.0	0.17	73	J5
3	0.69	6900	180	38.333	0.035	5.0	0.25	73	J5
4	0.73	7300	197.1	37.037	0.037	5.0	0.08	73	J5
5	0.77	7700	194	39.691	0.039	5.0	0.02	73	J5
6	1.7	17000	204.6	83.089	0.085	5.0	0.05	73	J5

Figure 7: SWMM (Pre) Model Input Assumptions

3. Post-LID Scenario (Porous Pavement):
    - Porous pavements were added to key subcatchments as LIDs.
    - Inputs for Subcatchment-1 included:
        - Surface Thickness: 100 mm
        - Void Ratio: 0.18
        - Seepage Rate: 482.6 mm/hr
        - Surface Slope: 0.2%
        - Impervious Area Treated: 60%.
        - % of Subcatchment Occupied: 21.3% and so on.

Results

The results of the SWMM simulations reveal the changes in surface runoff volumes under pre-development, post-development, and post-LID (porous pavement) conditions for various storm return periods (5-, 10-, 25-, and 50-year events). The findings demonstrate the significant impact of urbanization on runoff volumes and the effectiveness of porous pavements in mitigating these effects.

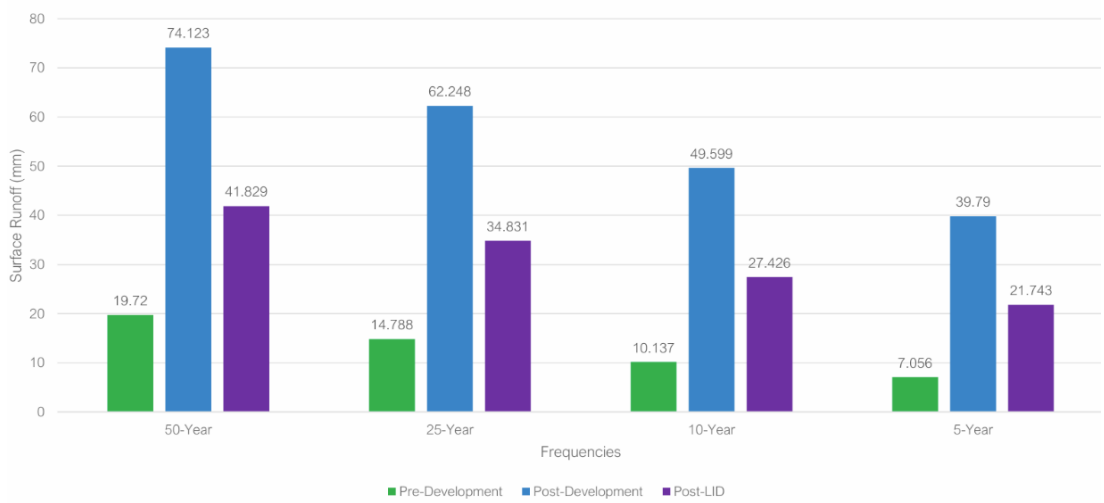

Figure 8: Comparison of surface runoffs in Pre, Post and Post-LID conditions

For the 50-year storm event, surface runoff increased from 19.72 mm in the pre-development scenario to 74.123 mm in the post-development scenario, highlighting the impact of impervious surfaces on stormwater flow. With the implementation of porous pavements, runoff was reduced to 41.829 mm, representing a 43.6% reduction compared to post-development conditions. Similar trends were observed for other storm events. For the 25-year storm, runoff decreased from 62.248 mm in the post-development case to 34.831 mm after introducing porous pavements, a 44.1% improvement. In the 10-year storm, runoff was reduced from 49.599 mm to 27.426 mm, a reduction of 44.7%, while for the 5-year storm, porous pavements brought runoff down from 39.79 mm to 21.743 mm, achieving a 45.4% reduction.

The results demonstrate a consistent pattern: urbanization significantly increases surface runoff due to the dominance of impervious surfaces, but porous pavements effectively reduce runoff volumes by enhancing infiltration. By implementing porous pavements, the strain on the stormwater system is alleviated, and peak runoff volumes are reduced, minimizing flood risks. Across all storm events, the reductions ranged between 43% and 45%, reflecting the effectiveness of porous pavements as a low-impact development strategy.

Conclusion

This study evaluated the impact of urbanization on stormwater runoff at Willow Elementary School and assessed the effectiveness of porous pavements as a low-impact development (LID) solution using SWMM simulations. The key findings are as follows:

- Urbanization caused a significant increase in surface runoff due to the dominance of impervious surfaces, leading to heightened flood risks and strain on the drainage system.
- The implementation of porous pavements reduced runoff volumes by 43% to 45%, effectively enhancing infiltration, alleviating flood risks, and improving system performance by mitigating peak flows and reducing pressure on existing infrastructure.
- Future research could focus on optimizing porous pavement designs to accommodate varying urban conditions, assessing their long-term durability and maintenance requirements, or integrating them with other LID solutions for holistic stormwater management.

In conclusion, porous pavements offer a sustainable and efficient solution for managing stormwater runoff in urbanized areas. By restoring natural infiltration processes, they provide an effective pathway for reducing runoff, mitigating flood risks, and enhancing the resilience of drainage systems in developed regions.

References

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SWMM Model (Pre) - Results

********** Runoff Quantity Continuity **********	5-year Volume hectare-m	5-year Depth mm	10-year Volume hectare-m	10-year Depth mm	25-year Volume hectare-m	25-year Depth mm	50-year Volume hectare-m	50-year Depth mm
Total Precipitation	0.545	60.167	0.660	72.793	0.802	88.520	0.933	102.951
Evaporation Loss	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Infiltration Loss	0.316	34.843	0.353	38.963	0.392	43.297	0.423	46.665
Surface Runoff	0.064	7.056	0.092	10.137	0.134	14.788	0.179	19.720
Final Storage	0.166	18.372	0.216	23.846	0.278	30.660	0.334	36.864
Continuity Error (%)	-0.173		-0.211		-0.253		-0.289	

SWMM Model (Post) - Results

********** Runoff Quantity Continuity **********	5-year Volume hectare-m	5-year Depth mm	10-year Volume hectare-m	10-year Depth mm	25-year Volume hectare-m	25-year Depth mm	50-year Volume hectare-m	50-year Depth mm
Total Precipitation	0.545	60.167	0.660	72.793	0.802	88.520	0.933	102.951
Evaporation Loss	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Infiltration Loss	0.093	10.310	0.102	11.223	0.110	12.137	0.116	12.815
Surface Runoff	0.360	39.790	0.449	49.599	0.564	62.248	0.672	74.123
Final Storage	0.094	10.369	0.112	12.351	0.132	14.611	0.150	16.581
Continuity Error (%)	-0.502		-0.521		-0.538		-0.552	

SWMM Model (Post-LID) - Results

********** Runoff Quantity Continuity **********	5-year Volume hectare-m	5-year Depth mm	10-year Volume hectare-m	10-year Depth mm	25-year Volume hectare-m	25-year Depth mm	50-year Volume hectare-m	50-year Depth mm
Total Precipitation	0.545	60.167	0.660	72.793	0.802	88.520	0.933	102.951
Evaporation Loss	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Infiltration Loss	0.283	31.184	0.333	36.808	0.395	43.650	0.451	49.804
Surface Runoff	0.197	21.743	0.248	27.426	0.316	34.831	0.379	41.829
Final Storage	0.068	7.487	0.080	8.867	0.094	10.426	0.107	11.779
Continuity Error (%)	-0.409		-0.424		-0.437			

SWMM RAINDATA				
t	i5	i10	i25	i50
1	19.236	23.183	28.279	32.737
2	13.212	15.978	19.436	22.594
3	10.605	12.852	15.608	18.188
4	9.074	11.012	13.358	15.593
5	8.040	9.768	11.839	13.839
6	7.284	8.857	10.727	12.552
7	6.700	8.154	9.869	11.559
8	6.232	7.589	9.181	10.762
9	5.847	7.124	8.614	10.105
10	5.522	6.732	8.137	9.551
11	5.244	6.396	7.728	9.076
12	5.003	6.104	7.373	8.663
13	4.790	5.848	7.060	8.300
14	4.602	5.620	6.783	7.977
15	4.433	5.415	6.534	7.688
16	4.280	5.231	6.310	7.427
17	4.142	5.063	6.106	7.190
18	4.016	4.910	5.921	6.974
19	3.900	4.770	5.750	6.775
20	3.793	4.640	5.593	6.592
21	3.694	4.520	5.447	6.422
22	3.602	4.408	5.311	6.264
23	3.516	4.304	5.185	6.117
24	3.436	4.207	5.067	5.979

Figure 9: Rainfall Data from NOAA AtlaS (Time is in hrs and Intensity in mm/hr)